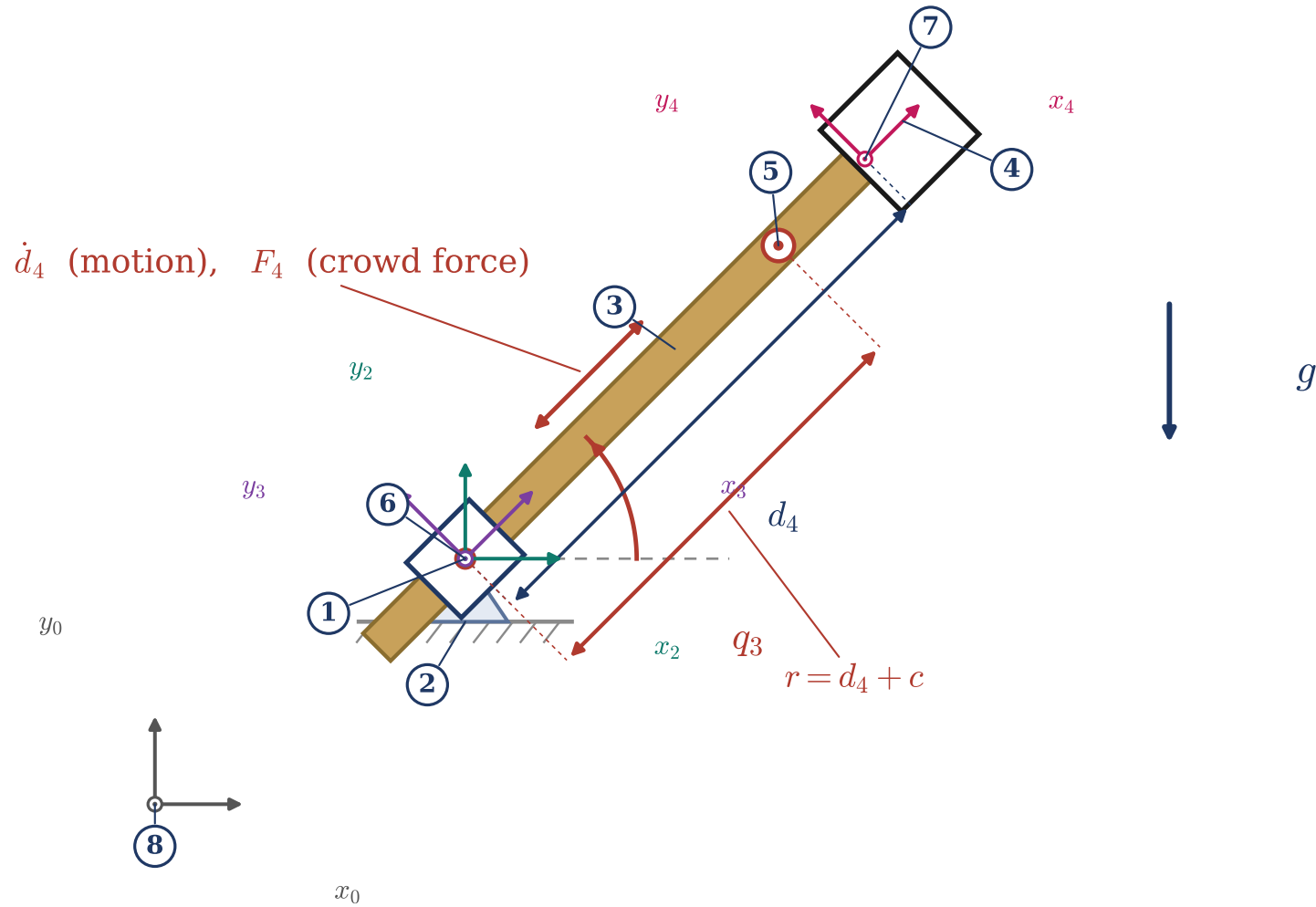


(b) Stage-1 abstraction: two degrees of freedom

q_3 saddle rotation (revolute) · d_4 crowd extension (prismatic) · boom welded to ground
swing and hoist-rope actuation enter at milestones M5–M6



Key

- 1 saddle pivot (revolute, q_3)
- 2 boom welded to ground (passive at Stage 1)
- 3 dipper handle (prismatic, d_4)
- 4 dipper
- 5 COM: $M_d = 54,300$ kg, $I_{zz} = 287,900$ kg m²
- 6 $\{o_2\} \equiv \{o_3\}$ saddle and crowd frames coincide ($a_3 = 0$)
- 7 $\{o_4\}$ handle-end frame
- 8 $\{W\}$ world frame

$r = d_4 + c$ is the pivot-to-COM lever arm;
 $c = -1.32$ m is a fixed bookkeeping constant.